

Xiao Chuyu

Robotics Engineering / Embedded Control / Medical Devices / Student Founder

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Third-year Robotics Engineering student at Chongqing University and Project Lead of Goodent. Across 20+ project-based builds, I connect mechanical design, embedded control, robotics, product validation and market evidence to create testable physical systems.



Education

Chongqing University (CQU) - B.Eng. in Robotics Engineering

Sep 2023 - Jun 2027
(Expected)

National School of Excellent Engineers | Mingyue Sci-Tech Innovation Class

- Academic record: GPA 3.68/4.0; Major Rank 9/58.
- International study: Selected through the university exchange programme for a funded Fall 2026 semester at the Singapore University of Technology and Design (SUTD).
- Languages: CET-4 534; CET-6 436; native Mandarin Chinese.

Engineering Skills

Mechanical	SolidWorks, Fusion 360, parametric CAD, DFM, 3D printing, assembly planning
Simulation	COMSOL Multiphysics, ADAMS, PSIM, MATLAB
Embedded	STM32, ESP32-S3, Arduino, PCB, C/C++, sensor integration, motor control
Robotics	LeRobot, ACT, ArUco, MediaPipe, teleoperation, demonstration datasets
Product	Prototype iteration, user research, competitive analysis, verification planning, pitching

Selected Project Experience

Goodent Intelligent Dental Power System

2026 - Present

Project Lead / Co-founder

- Lead product definition, system architecture, motor control and team delivery for a clinically oriented intelligent dental power system.
- Advanced three prototype generations toward engineering validation; building the torque, speed, thermal and abnormal-load test chain.
- Secured RMB 500k in seed funding and began medical-device regulatory preparation.

LeRobot Dental Robotics

2026

System Integration / Vision Calibration / Dataset Analysis

- Built a dental-manipulation workflow on a 6-DOF arm using ArUco pose alignment, leader-follower teleoperation, LeRobot and ACT.
- Reviewed dataset distributions, joint trajectories, smoothing behavior and task keyframes to improve demonstration quality.

DIY BLDC Motor - Coreless V1 to Inner-Rotor V2

2026

Motor Mechanical Design / Winding / Embedded Drive

- Built and ran a 12-slot/16-pole coreless BLDC prototype with a star winding, ESP32-S3 and sensorless ESC.
- Carried winding, phase-order, magnet-layout and mechanical-support lessons into a compact inner-rotor V2 design.

Gamma-Type Stirling Engine

2025

Mechanical System Design / Simulation / Prototype Fabrication

- Completed a 41-part parametric assembly and connected thermodynamic analysis, COMSOL multiphysics, ADAMS dynamics and physical prototyping.

Work, Venture and Leadership Experience

Zhixing Technology <i>Co-founder / Product Design Intern</i> - Contributed to parametric off-road vehicle modelling, part-library management, industrial design and competitor research with SolidWorks and Creo. - Supported product and venture execution as a co-founder; the project secured seven-figure RMB angel financing.	2024
Lanjingling Smart Planter - National Innovation Programme <i>Project Lead</i> - Led user needs, enclosure design, sensing, automatic irrigation, embedded integration and two prototype iterations. - Completed project documentation and defense; received an Excellent national-project completion rating.	2025
CQ-U Robocon Team and Course Teaching <i>Robocon Member / Teaching Assistant</i> - Contributed robot chassis and suspension design while practising system decomposition and cross-functional collaboration. - Supported Ergonomics and Robotics Basics courses as a teaching assistant.	2025

Additional Engineering Projects

3D-Printed Pneumatic Vehicle <i>Vehicle Mechanical Design / Transmission / DFM</i> Decomposed the chassis, engine, gearbox, differential, steering and suspension into printable, serviceable modules backed by a complete BOM.	2025
ECG Signal Processing and Embedded Acquisition <i>DSP / Embedded C / MATLAB</i> Developed filtering, QRS detection and heart-rate analysis workflows across MATLAB and embedded C, including GUI-based signal inspection.	2025
Human Mesh Strain Analysis <i>Python / Trimesh / Blender</i> Built a mesh-deformation workflow to compare body poses using area strain, shear and edge-length metrics with vertex-color visualization.	2026

Selected Coursework and Engineering Methods

Coursework	Automatic Control, Microcircuit Design, Numerical Analysis, Engineering Design, Ergonomics, Robotics Basics
Motor Systems	Back-EMF and commutation analysis, winding and pole review, layered STM32 motor-control debugging
Product Validation	User context to verifiable requirements, prototype-to-EVT stage gates, medical-device risk-to-test planning
Market Practice	User interviews, competitive positioning, venture evidence, pitch and fundraising preparation

Selected Recognition and Evidence

- Goodent: RMB 500k seed funding; three prototype generations; engineering validation in progress.
- Zhixing venture: Project secured seven-figure RMB angel financing.
- National Innovation Programme: Excellent completion rating as project lead.
- SUTD: Funded Fall 2026 exchange placement selected independently through the university programme.
- Portfolio: 20+ project-based builds spanning mechanics, simulation, embedded control, robotics and product development.

Project evidence, prototype media and engineering notes are available at lithos666.github.io.